

MACHINE LEARNING IN CYBERSECURITY · SPRING 2026

Machine Learning for IoT Network Intrusion Detection

Binary & Hierarchical Multi-Class Classification with Explainability and Adversarial Robustness

M M Nishat · April 2026

The Growing IoT Attack Surface

Billions of IoT devices generate continuous network traffic and most lack native security. The result is a massive, constantly expanding attack surface that demands that demands automated, real-time detection.



IoT Devices

Traffic Capture

ML Classifier

Security Alert

This project builds ML classifiers that detect DDoS attacks with **>99% accuracy**, classify 34 IoT attack subtypes into 7 families, and explain decisions using SHAP while measuring adversarial fragility.

DATASET

CIC IoT Dataset 2023

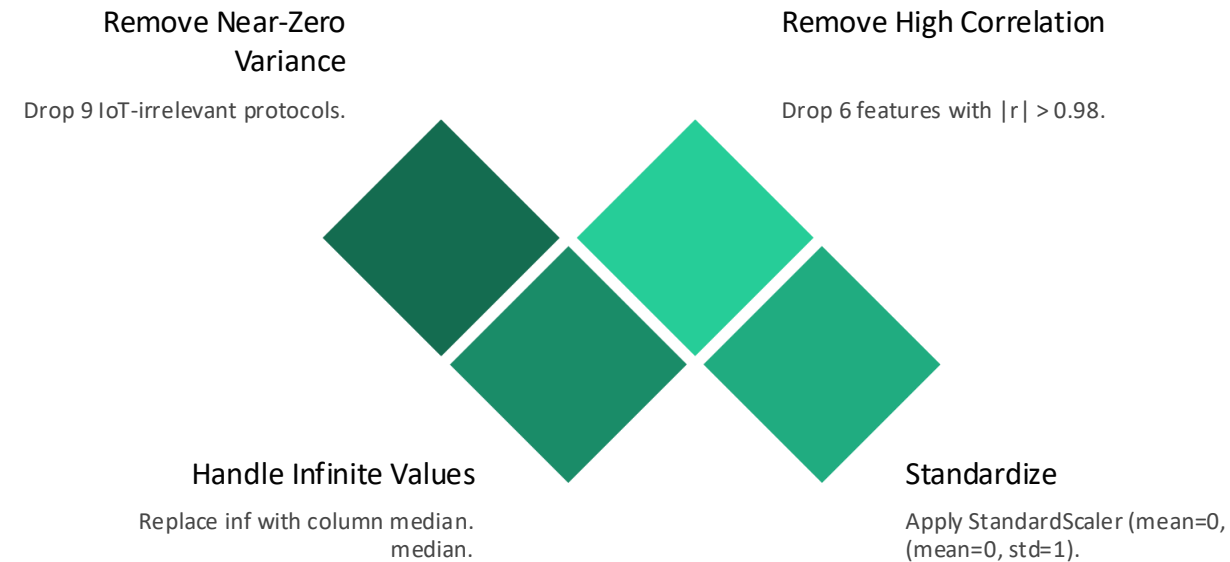
Dataset at a Glance

- **Source:** Canadian Institute for Cybersecurity
- **508,155** total instances
- **34** attack subtypes
- **46** network flow features
- Binary subset: 36,000 rows (18K Benign vs. 18K DDoS-HTTP_Flood) - perfectly balanced
- Multi-class: 34 subtypes grouped into **7 families**

7 Attack Families

- **DDoS** - Distributed denial-of-service
- **DoS** - Single-source denial-of-service
- **Mirai** - IoT botnet malware
- **Recon** - Network scanning & enumeration
- **Spoofing** - Identity & address forgery
- **WebAttack** - HTTP-layer exploits
- **Benign** - Normal traffic baseline

Data Preprocessing & Feature Selection



Each step reduces noise and redundancy, preparing clean input for all five models.

Why These Steps Matter

- Infinite values replaced with column median to prevent model divergence
- Infinite values replaced with column median to prevent model divergence
- 9 near-zero variance features removed (SSH, SMTP, IRC - absent in IoT traffic)
- 6 highly correlated features dropped ($|r| > 0.98$) to reduce multicollinearity
- StandardScaler applied: mean=0, std=1 for gradient-based learners

Final Feature Counts

31 Features

Binary classification task

37 Features

Multi-class classification task

Five Algorithms, One Rigorous Pipeline



Logistic Regression

Linear baseline - interpretable sanity check check



Random Forest

200 trees - robust to noise



XGBoost

Gradient boosting - industry standard for tabular data



LightGBM

Histogram boosting - fastest training, handles handles high-dim data



LSTM

2-layer deep learning - captures sequential flow patterns



Experimental Setup: Three holdout splits (60/40, 70/30, 80/20) + 10-fold stratified CV.

Metrics: Accuracy, Precision, Recall, F1.

Binary Classification: Near-Perfect Detection

Model	Accuracy	Precision	Recall	F1
Logistic Reg.	98.36%	99.00%	97.72%	98.35%
Random Forest	99.98%	100.0%	99.96%	99.98%
XGBoost	99.97%	99.98%	99.96%	99.97%
LightGBM	99.98%	99.99%	99.97%	99.98%
LSTM	99.88%	99.94%	99.81%	99.87%

Key Takeaways

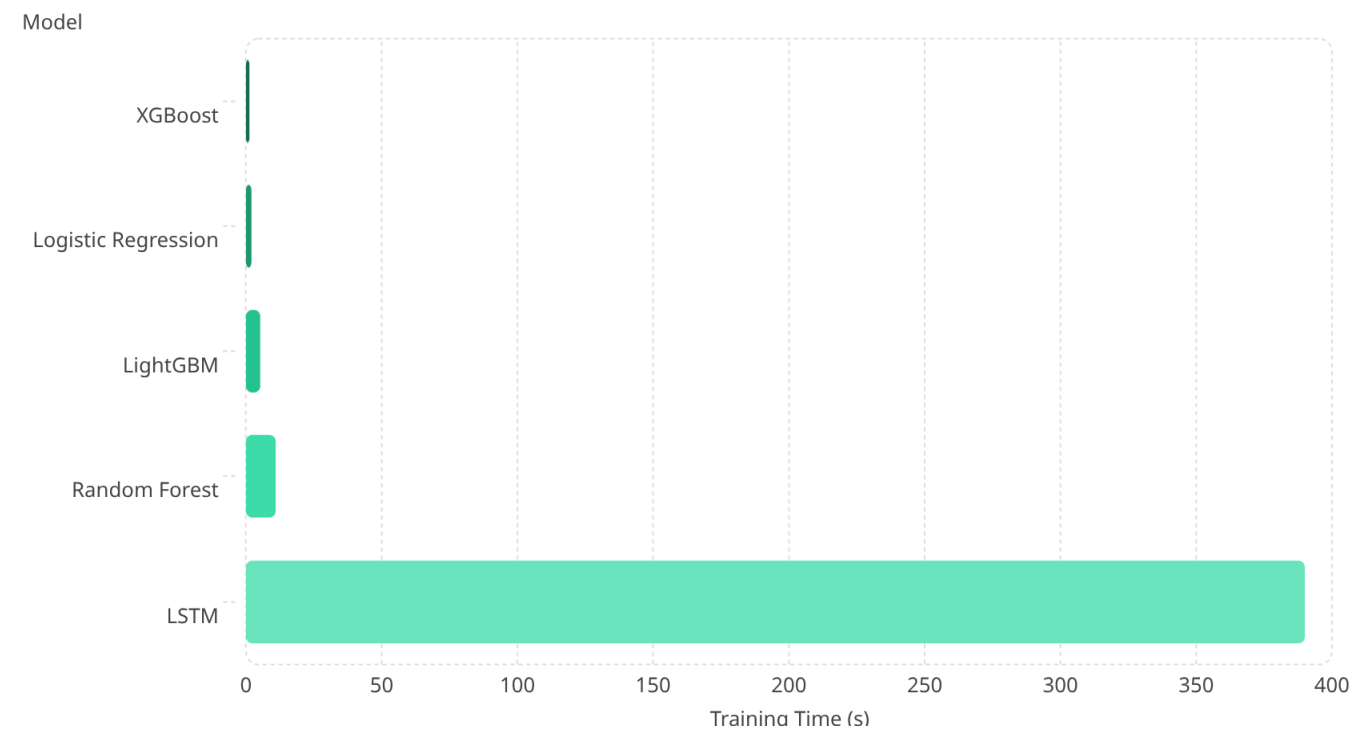
- Tree-based models all achieve **≥99.97% F1** on 10-fold CV
- Best overall: **Random Forest & LightGBM** (tied at 99.98%)
- Logistic Regression: 98.35% - slightly non-linear decision boundary
- LSTM: 99.87% on CV, but **unstable on holdout splits** due to early stopping sensitivity



LSTM Split Instability: 99.88% (60/40) → 81.64% (70/30) → 86.57% (80/20). Early stopping triggered after only 8 epochs - 10-fold CV is the reliable metric.

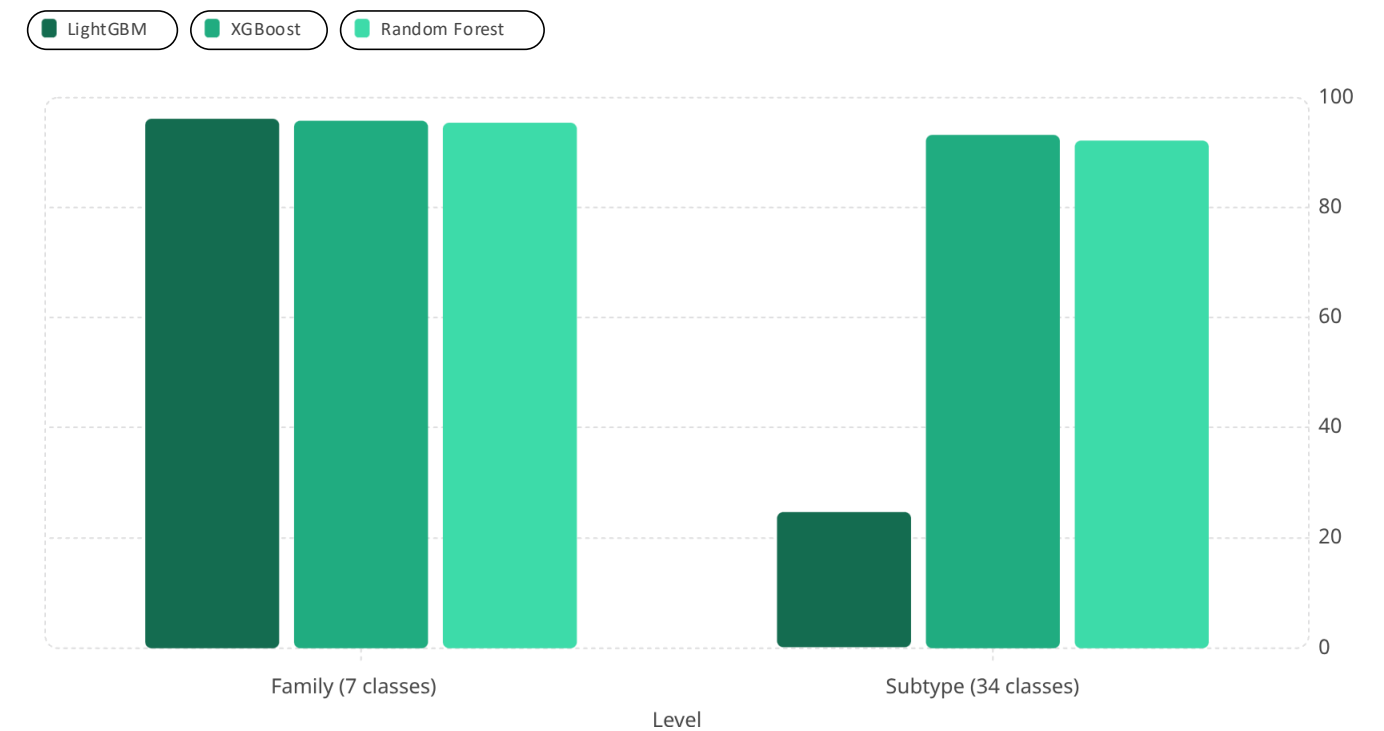
Training Time & Multi-Class Results

Training Time Comparison



Takeaway: XGBoost offers the best speed-accuracy tradeoff - under 2 seconds vs. 390s for LSTM.

Hierarchical Multi-Class Accuracy



LightGBM's histogram binning collapses at 34-class granularity (24.59%).
XGBoost leads subtype-level at 93.22% accuracy, 87.88% F1-macro.

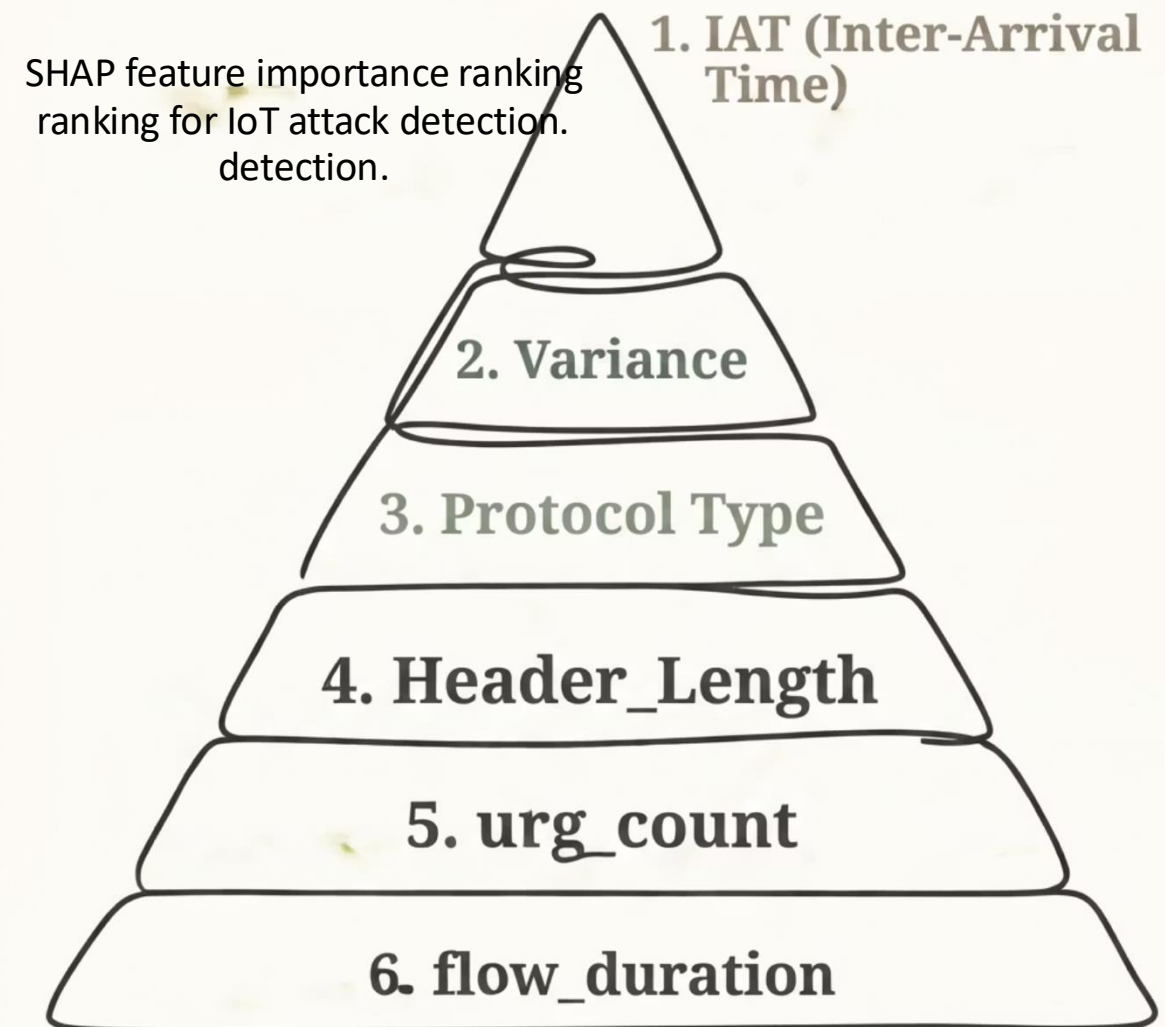
SHAP Explainability

What SHAP Reveals

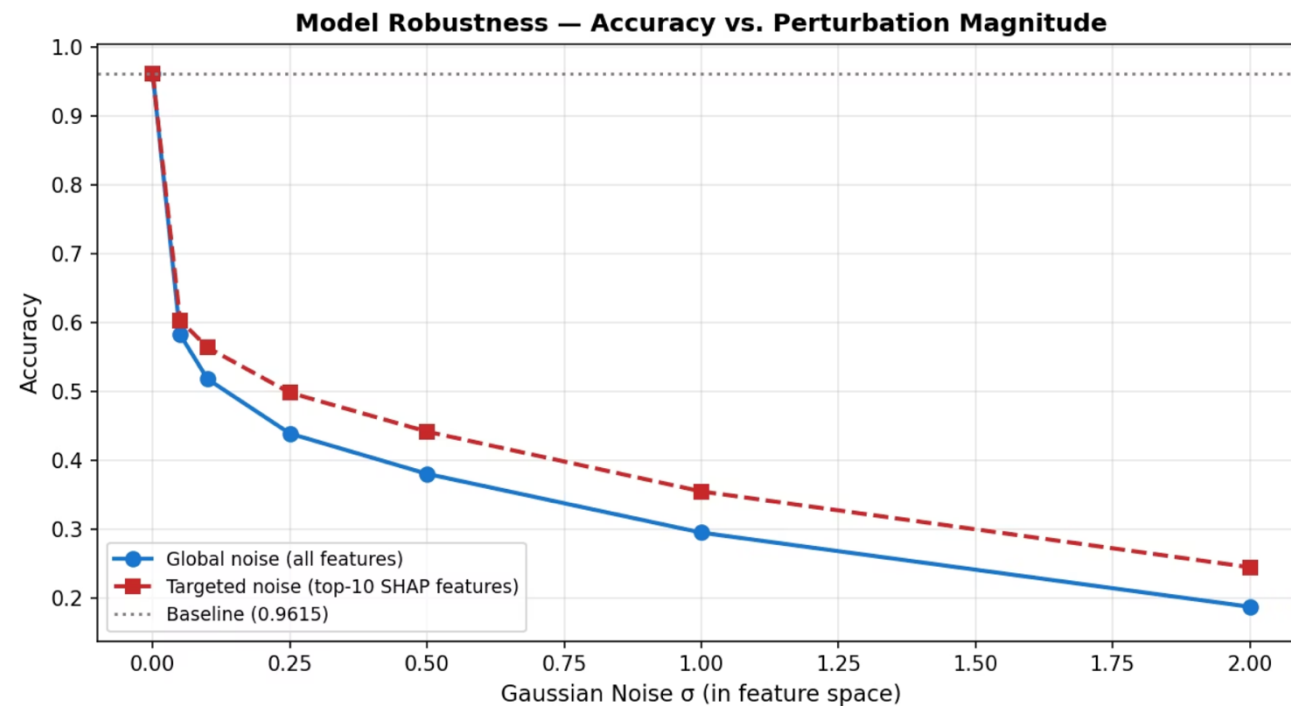
SHAP TreeExplainer on the LightGBM family classifier identifies **Inter-Arrival Time Arrival Time (IAT)** as the dominant signal - its SHAP value of **3.08** is **5× higher** than **higher** than the #2 feature.

- **DDoS:** IAT, flow_duration
- **Mirai:** Protocol Type, Header_Length
- **WebAttack:** Variance, Duration
- **Recon:** urg_count, IAT

Per-family feature importance enables **targeted firewall rule construction** - a **construction** - a direct path from model insight to deployment action.



Adversarial Robustness: Fragile Under Noise



What the Numbers Tell Us

- Even $\sigma=0.05$ drops accuracy from 96.15% $\rightarrow$ 58.26% (**-37.89%**)
- Zeroing just **IAT** drops accuracy by **41.7%** — confirming its centrality
- **Most robust:** Mirai (only 10.98% drop at $\sigma=0.5$) — distinctive protocol patterns
- **Most fragile:** DoS (92.06% drop) and Benign (81.58% drop)

⊗ Small Gaussian noise causes major degradation. **Adversarial training** is essential before real-world deployment.

Key Findings & Real-World Impact

Best Binary Model

Random Forest / LightGBM - F1 = **99.98%**

Best Multi-Class

XGBoost - **93.22%** on 34 subtypes

Top Feature

IAT - **5×** more important than #2

Robustness Gap

Mirai: **10.98%** drop vs. DoS: **92.06%** drop

Real-World Applications

- IoT gateway DDoS detection at the edge
- SOC alert triage via hierarchical classification
- SHAP-driven firewall rule construction
- Robustness-aware deployment planning
- Rapid retraining - XGBoost under 2 seconds

Limitations & Future Work

- LSTM split-sensitive - CV is essential
- LightGBM histogram binning fails at 34-class granularity
- Adversarial training needed to close robustness gap
- Future: full CICIOT2023 dataset, real IoT testbed deployment

References

- Neto, E. C. P., Dadkhah, S., Ferreira, R., Zohourian, A., Lu, R., & Ghorbani, A. A. (2023). CICIOT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment. *Sensors*, 23(13), 5941.
- Chen, T., & Guestrin, C. (2016, August). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference international conference on knowledge discovery and data mining* (pp. 785-794).
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems*, 30.
- Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems, processing systems*, 30.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural computation*, 9(8), 1735-1780.
- Hosmer Jr, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression*. John Wiley & Sons.
- Breiman, L. (2001). Random forests. *Machine learning*, 45(1), 5-32.

Questions?